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Thick roots and less microaggregates improve hydrological connectivity

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HIGHLIGHTS

- Quantifying the effect of root-soil complex on hydrological connectivity.
- Abundant thick roots and less microaggregates improve the hydrological connectivity.
- Microaggregates dominated in effect of root-soil complex on hydrological connectivity.
- Hydrological connectivity of wetland with high water content is more difficult to assess.

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ABSTRACT

Due to human activity and global climate change, the Yellow River Delta, the youngest delta wetland in China, is suffering serious degradation. The study of hydrological connection provides new perspectives and technical support for the protection and restoration of delta wetlands. To quantify the interaction between the hydrological connection and the root–soil complex, the current study took dye-tracing experiments to examine the small-scale hydrological connectivity in soil where *Phramites australis* grew. The dye coverage was selected as the indicator of hydrological connectivity after preliminary analysis in this study. The main results were that (1) the strength of hydrological connectivity was negatively related to the microaggregates content, but had little to do with other soil physical properties; (2) there was a notable positive correlation between the indexes of thick root (D > 5 mm) and the dye coverage hydrological connectivity, while root biomass had little effect on hydrological connectivity; and (3) the influence of the microaggregate content dominated in the combined effect of the total surface area of the root (D > 5 mm) and the microaggregate content on hydrological connectivity in each soil layer.

1. Introduction

Wetland loss resulting from climate change and the enormous demand for natural resources with society developing is a global problem (Guan et al., 2013; Thorne et al., 2018; Xiao et al., 2019). Wetland protection and restoration is extremely important because of the increasing seriousness of wetland degradation (Cui et al., 2018; Han et al., 2006; Ma et al., 2017). Hydrological processes are the foundation on which wetlands are based and these processes has direct influences on biological processes (Li et al., 2009). An in-depth understanding of wetland hydrology can help with understanding the reasons behind wetland degradation. It can also help in the formulation of appropriate protection and restoration measures (Liu et al., 2019a). Indeed, the mechanisms between hydrology, vegetation, and soil in wetland ecosystems are considered to be the key to wetland ecosystem process (Liu et al., 2018a, 2019a, 2020b; Zhao et al., 2016).

The original definition of hydrological connection was put forward in the early 1980s (Vannote et al., 1980) and at the beginning,

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referred to the geological connection between water on the ground (Hooke 2003). This provided a new perspective for studying the interactions between hydrology and other ecosystem factors such as the biota (Liu et al., 2020a; Wu et al., 2020). Hydrological connectivity has been investigated at various scales, which range from the global scale to the soil column scale (Bai et al., 2012; Dai et al., 2020b; Freyberg et al., 2014; Liu et al., 2018b, 2019b, 2020a; Miyata et al., 2019). For instance, the hydrological connection among hillslopes and riparian zones has been investigated, and it has been shown that hydrological connectivity can influence the composition of the streamflow (Freyberg et al., 2014). Also, the hydrological connectivity at the soil column scale has been shown to be spatially heterogeneous as a result of changes in soil properties (Dai et al., 2020b). However, attentions of most researchers have been concentrated on that how hydrological processes impact animals, plants, and soil (Freyberg et al., 2014; Hooke 2003; Levia et al., 2011; Liu et al., 2020a; Wu et al., 2020). The interaction between these elements is mutual: the hydrological processes affect the soil composition as well as the growth and distribution of animals and plants (Liu et al., 2018a, 2019b, 2020b; Rosier et al., 2015, 2016), and the animals and plants affect the hydrological connectivity, which in turn affects the hydrological processes (Hinsinger et al., 2009; Jarvis et al., 2007; Stan et al., 2020; Weiler and McDonnell 2007; Zhang et al., 2015b). The focus of earlier studies was on the relationship between the soil and hydrological processes or the relationship between root systems and hydrological processes, respectively; however, in recent years, studies of small-scale hydrology have begun to take the root−soil complex as the object of their research (Johnson and Jost, 2011; Scheu et al., 2003; Stan et al., 2020). In this study, the “root-soil complex” is the mobile colluvium layer which combines roots, soil and interface elements (Fig. 1), which contains the rhizosphere. In practice, the area that has an influence on the hydrological processes extends beyond the rhizosphere, which means that the hydrological connectivity at this scale is not just affected by the soil or root system but by the entire root−soil complex. Therefore, it is necessary to study the relationship between the root−soil complex and hydrological connectivity at the small-scale.

In the literature, the hydrological connectivity is classified as being either horizontal (which includes lateral hydrological connectivity and longitudinal hydrological connectivity) or vertical (Merenlender and Matella 2013). Vertical hydrological connectivity refers to the hydrological connectivity from top to bottom or bottom to top in the vertical direction, such as the connectivity between surface water and groundwater (Merenlender and Matella 2013). This study mainly focused on the connectivity of flow in soil, which is a kind of underground vertical hydrological connectivity (Dai et al., 2020b). Underground vertical hydrological connections are also very important: these can directly affect the properties of the soil as well as the development of plants through their root systems; they can even affect the microorganisms and animals found in the soil (Rosier et al., 2015, 2016; Stan et al., 2020). Nutrients and even microorganisms are constantly exchanged between places that have a strong vertical hydrological connectivity; this exchange also has an effect on the microenvironment (Rosier et al., 2016) and means that the biological activity in such areas is very intense and frequent. It can be seen that measurements of the small-scale hydrological connectivity could be made to assess the biological activity in a particular area, and this would provide a relatively simple and intuitive understanding of the ecological status of the area (Mooney and Korošak 2009; Seyfarth et al., 2012; Wang and Zhang 2011; Zhang et al., 2017). Compared with other experimental methods, a dye-tracing experiment is simpler and more intuitive: the hydrological processes can be directly observed after a few treatments, and the experiment is also simple to carry out and does not require a lot of resources (Wang and Zhang 2011).

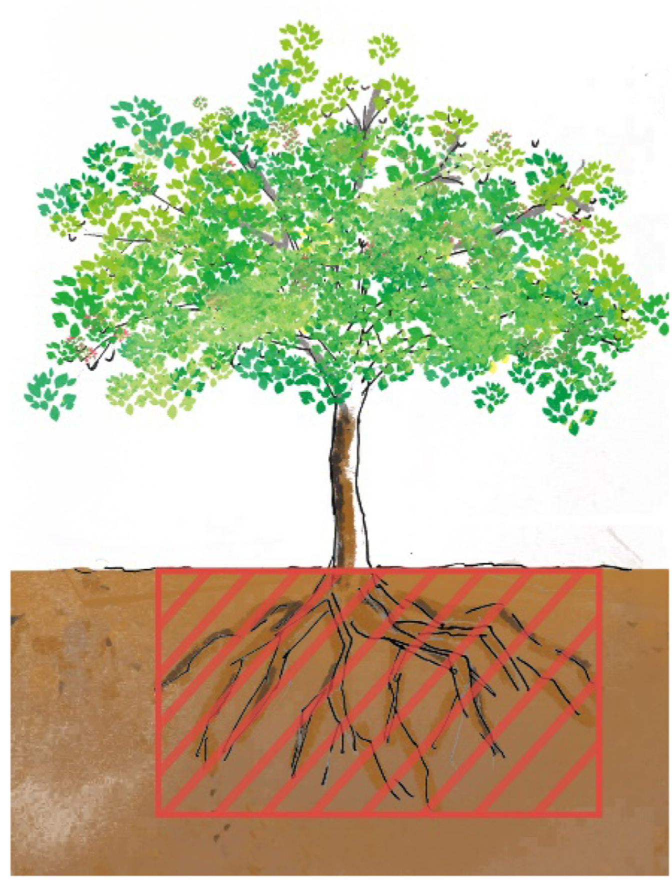

Fig. 1. Schematic diagram of the root−soil complex (The area enclosed by the red box is the approximate area of the root−soil complex.). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

In this study, we took an area where *Phragmites australis* was found as the focus of our research and aimed to explore the relationship between the small-scale hydrological connectivity and the environmental elements (plant roots and soil) through simple and intuitive dye-tracing experiments. We also wanted to explore the relationship between the small-scale hydrological connectivity and the root−soil complex. The results of this study provide a new method for determining the degree of wetland degradation and can also be used to guide the protection and restoration of wetlands.

2. Materials and methods

2.1. Fieldwork

We selected three different plots (Pc1, Pc2, and Pc3) from representative *Phragmites australis* communities that were randomly distributed within the wetland and where the ground was flat. The plots were located at 37°48′44″N, 118°58′57″E at a height of 3.0 m a.s.l. (Fig. 2) and were separated by more than 3 km. To insure the accuracy of the results, we waited until there had been no rain for at least seven days before the fieldwork. All weeds and litter were also carefully removed from the plots for minimizing the interference to the surface. We collected soil samples from each plot and kept the samples sealed, then making measurements of their properties (Table 1).

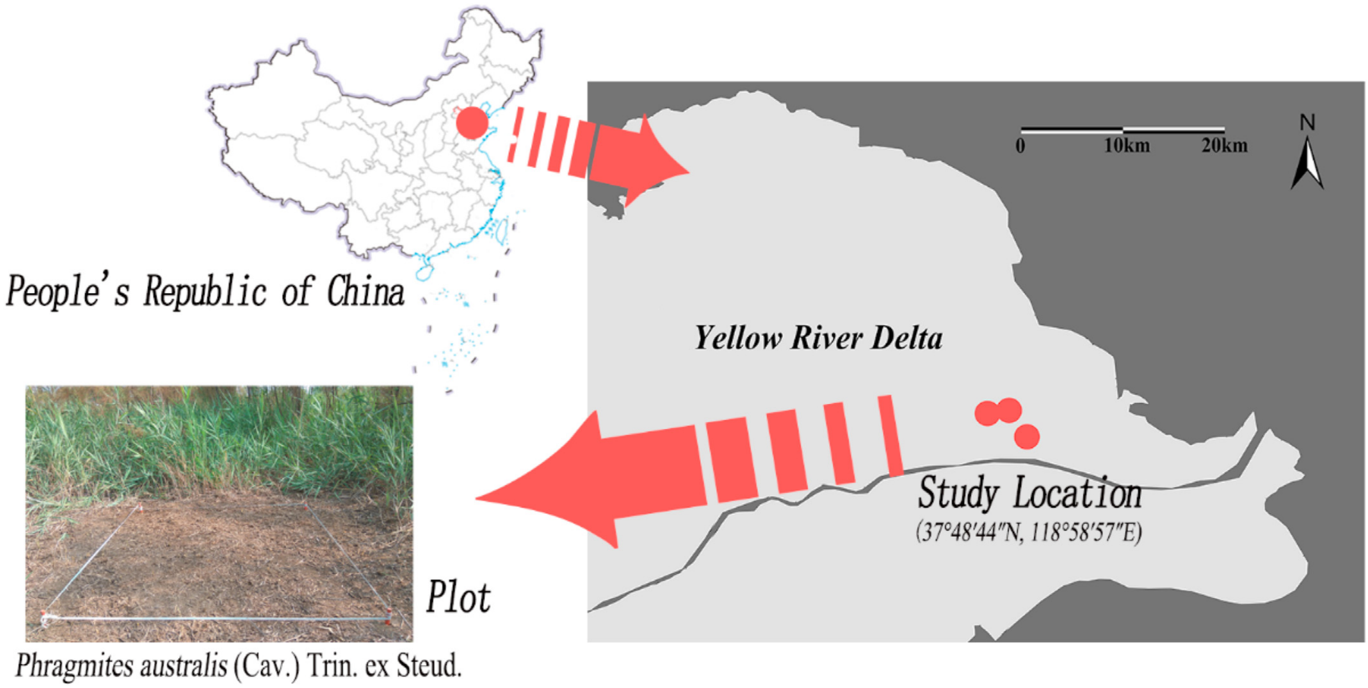

Fig. 2. Location of the study area and plot after pretreatment.

Table 1
Physical properties of the soils found at the measurement plots.

Soil physical properties	Plot no.		
	Pc1	Pc2	Pc3
Mass water content (MWC)	0.213 ± 0.042	0.206 ± 0.007	0.237 ± 0.099
Bulk density (BD)	1.386 ± 0.035	1.413 ± 0.051	1.366 ± 0.780
Total porosity (TP)	0.467 ± 0.016	0.471 ± 0.003	0.486 ± 0.260
Capillary porosity (CP)	0.461 ± 0.018	0.466 ± 0.005	0.482 ± 0.256
Maximum moisture capacity (MaxMC)	0.337 ± 0.019	0.334 ± 0.015	0.357 ± 0.184
Capillary moisture capacity (CMC)	0.332 ± 0.018	0.325 ± 0.011	0.347 ± 0.181
Minimum moisture capacity (MinMC)	0.321 ± 0.020	0.312 ± 0.007	0.333 ± 0.174
Soil water storage (SWS)	0.031 ± 0.006	0.030 ± 0.002	0.034 ± 0.014
Soil permeability (SP)	0.429 ± 0.009	0.434 ± 0.003	0.443 ± 0.243
Lower limit of the optimum moisture content (LLOPC)	0.225 ± 0.014	0.218 ± 0.005	0.234 ± 0.122
Microaggregate content (MC)	98.382 ± 3.234	99.564 ± 0.866	99.225 ± 0.765
Sand content (SaC)	27.107 ± 17.145	49.848 ± 9.701	28.122 ± 20.823
Silt content (SiC)	58.274 ± 17.027	38.173 ± 10.268	56.650 ± 18.712
Clay content (ClC)	14.619 ± 4.343	11.980 ± 2.081	15.228 ± 3.046

2.2. Dye-tracing experiments

We chose an appropriate place where reeds (*Phragmites australis*) grew well within every plot to set up a quadrat of which size is 1.2 m × 1.2 m. We then applied dye solution (C.I. Food Blue 2) to the surface of each quadrat evenly. The solution had a concentration of 5 g/L. The total volume of the solution applied to each quadrat was 50 L. After the application of the dye had been completed, plastic film was covered the sample site for keeping out disturbance and to stop evaporation from the surface. Twenty-four hours after the application of the dye solution, in order to avoid the boundary effect, we selected an area measuring 1.0 m × 1.0 m from the central part of the quadrat as the sampling area, and then excavated the soil along vertical profiles within this area.

First, we excavated the vertical profiles (Fig. 3). The interval between slices was 20 cm (for a more complex dye pattern, this distance could be reduced). Seven vertical profiles were excavated finally. After each profile had been prepared, we acquired dye patterns by photos. The excavation depth depended on the depth to which the roots had grown. Pictures were then transformed into gray-scale images and binarized to obtain black-and-white biphasic pictures by using MATLAB software. Finally, Image-Pro Plus was used to produce a vectorized black-and-white biphasic map (Fig. 4) (Dai et al., 2020a).

After applying a large amount of dye solution, we could get lots of soil profile indexes reflecting the possible water movement through the analysis of dye patterns. And those indexes were indicators of hydrological connectivity in root–soil complex.

2.3. Soil profile indexes

These indexes were results from the analysis of dye pattern images, and they were indicators of the hydrological connectivity in root–soil complex.

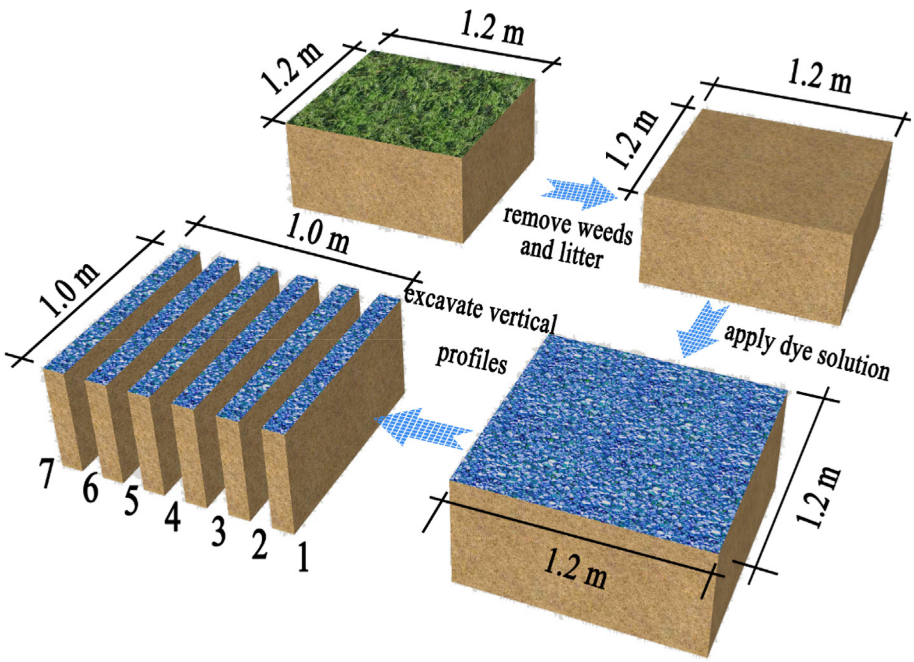

Fig. 3. Schematic diagram of experience design (The numbers represent vertical soil profiles from one side to the other side.).

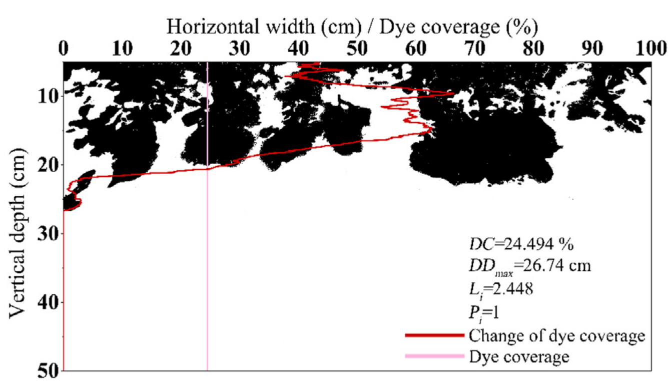

Fig. 4. Example of image processing and analysis (Where *DC* is Dye coverage, DD_{max} is the maximum dyed depth, the red curve represents the change of dye coverage with depth, and the pink line represents the total dye coverage of the profile.). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

2.4. Dye coverage

Any change in the dye coverage within the profile reflects a change in the movement of water, which, in turns, means that there has been a change in the strength of the small-scale hydrological connectivity. The dye coverage can be calculated as

$$DC = \frac{D}{D + ND} \times 100\% \tag{1}$$

where *DC* is the total dye coverage of each profile (%), *D* is the area which is dyed (cm^2), and *ND* is the area which is undyed (cm^2).

2.5. Maximum dyed depth

It is the depth to which the dye could reached deepest (cm).

2.6. Length index

The length index (L_i, %) is the sum of the absolute values of the difference between every two adjacent layers within each vertical profile; it can be calculated as

$$L_i = \sum_{i=1}^{n} |D_{c(i+1)} - D_{ci}| \tag{2}$$

Here, D_{ci} (%) represents the dye coverage of layer i within the profile while $D_{c(i+1)}$ (%) represents that of layer $i + 1$; and n is the total number of layers which is seven in this paper.

2.7. Peak value

P_i (peak value) is the total number of intersections between the straight vertical line (the number of intersections between red curve and pink line in Fig. 4) represents the line of total dye coverage value of profile and the graphic curve of the dye coverage changing with depth (intersections between red curve and pink line in Fig. 4).

3. Results

3.1. Dyeing indexes for the soil profiles

It can be seen from Table 2 that there is a general decrease in the values of the different dyeing indexes: the exceptions are the length index and the peak value. The length value is largest for Pc1 and smallest for Pc2; the peak value is largest for Pc3 and smallest for Pc1. At all the plots, the dye coverage decreases as the soil depth increases (Fig. 5).

3.2. Soil physical properties and dyeing indexes of soil profiles

In order to simplify the calculations and analysis, we selected some of the more important indicators by analyzing the correlation between the physical properties of the soil (Table 3). We then

Table 2
Dyeing indexes of soil profiles at the field plots.

Plot no.		Pc1	Pc2	Pc3	Average
Dye coverage (%)		20.68 ± 1.764	19.55 ± 3.336	17.83 ± 1.290	19.35 ± 1.288
Maximum dyed depth (cm)		33.93 ± 3.998	37.13 ± 3.062	34.41 ± 2.988	35.15 ± 1.878
Dye coverage in layer (%)	0–10 cm	10.70 ± 0.836	8.71 ± 0.821	8.38 ± 0.440	9.26 ± 0.458
	10–20 cm	8.46 ± 0.751	8.15 ± 1.376	7.17 ± 1.008	7.93 ± 0.602
	20–30 cm	1.38 ± 0.293	2.24 ± 1.123	2.13 ± 0.398	1.92 ± 0.397
	30–40 cm	0.04 ± 0.025	0.40 ± 0.236	0.11 ± 0.049	0.18 ± 0.084
	40–50 cm	0.10 ± 0.083	0.06 ± 0.036	0.04 ± 0.042	0.07 ± 0.032
Length index		2.62 ± 0.132	2.40 ± 0.183	2.43 ± 0.079	2.48 ± 0.079
Peak value		1.57 ± 0.571	1.86 ± 0.404	3.29 ± 1.107	2.24 ± 0.447

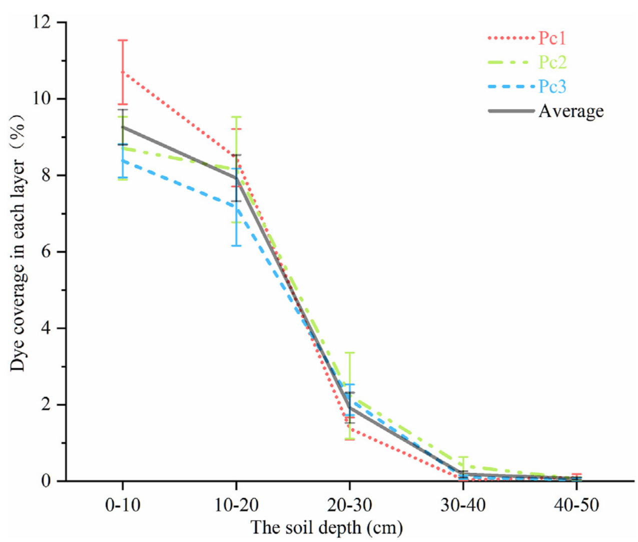

Fig. 5. The amount of dye coverage in each layer.

selected nine indexes that had no significant correlation between them. These were the bulk density, total porosity, capillary moisture capacity, minimum moisture capacity, soil water storage, microaggregate content, sand content, silt content, and clay content.

We then conducted a correlation analysis for the soil physical properties and the dyeing indexes of the soil profiles (Table 4). It was found that the only significant positive correlation was between the dye coverage for each layer and the microaggregate content; the only significant negative variation was between the dye coverage and the total porosity. There was no significant correlation between any of the other physical properties and the dyeing indexes. These results mean that when the microaggregate content in the soil increases, the dye coverage in each layer decreases and that when the total porosity of the soil increases, the coefficient of variation for the dye patterns increases.

3.3. Plant root indexes and dyeing indexes of the soil profiles

From Table 4, it can be seen that there is a significant correlation between the dye coverage of each layer and the plant root indexes (for roots which have a diameter larger than 5 mm; i.e., D > 5 mm). The plant root indexes include the total length, total width, total surface area, total projected area, and total volume. There is also a significant correlation between the coefficient of variation and the total surface area, total projected area, and total volume of the plant roots (for D > 5 mm). This means that as the total length, total width, total surface area, total projected area, or total volume of the plant roots increases (for D > 5 mm), the amount of dye coverage in each layer becomes smaller; however, the coefficient of variation then increases. In addition, it can be seen that when the total surface area or total projected area of the plant roots (for 1 mm < D < 3 mm) increases, the maximum dyed depth becomes smaller; the length index may also increase as the total volume increases.

3.4. Root–soil and dyeing indexes of soil profiles

From the earlier experiments, it had been found that microaggregate content and plant root indexes (D > 5 mm) were both significantly correlated with the dye coverage of each layer. We

Table 3
Correlation analysis for the soil physical properties.

	MWC	BD	TP	CP	MaxMC	CMC	MinMC	SWS	SP	LLOPC	MC	SaC	SiC	ClC
MWC	1.000													
BD	−0.451	1.000												
TP	0.845**	−0.575	1.000											
CP	0.898**	−0.509	0.987**	1.000										
MaxMC	0.705**	−0.904**	0.867**	0.819**	1.000									
CMC	0.750**	−0.874**	0.885**	0.849**	0.987**	1.000								
MinMC	0.792**	−0.814**	0.898**	0.872**	0.950**	0.983**	1.000							
SWS	0.966**	−0.206	0.755**	0.831**	0.508	0.566	0.631*	1.000						
SP	0.616*	−0.648*	0.938**	0.879**	0.883**	0.875**	0.859**	0.481	1.000					
LLOPC	0.792**	−0.812**	0.899**	0.874**	0.949**	0.983**	1.000**	0.632*	0.861**	1.000				
MC	−0.060	−0.289	0.203	0.190	0.258	0.295	0.324	−0.15	0.347	0.324	1.000			
SaC	−0.109	0.385	−0.254	−0.213	−0.361	−0.42	−0.438	0.001	−0.332	−0.443	−0.220	1.000		
SiC	0.133	−0.398	0.264	0.233	0.377	0.438	0.451	0.018	0.332	0.456	0.226	−0.988**	1.000	
ClC	−0.120	−0.015	0.006	−0.071	−0.006	−0.009	0.027	−0.115	0.085	0.034	0.018	−0.323	0.173	1.000

* the correlation is significant at the 0.05 level (two-tailed).
** the correlation is significant at the 0.01 level (two-tailed).

Table 4
Analysis of the correlation between the dyeing indexes of the soil profiles and the soil physical properties.

Soil property	Dye coverage of each layer	Maximum dyed depth	Length index
Bulk density	−0.198	0.533	−0.202
Total porosity	−0.185	0.255	−0.578
Capillary moisture capacity	−0.057	−0.255	−0.101
Minimum moisture capacity	−0.12	−0.378	0.029
Soil water storage	0.031	−0.098	−0.257
Microaggregate content	−0.621*	0.981	−0.987
Sand content	0.442	0.864	−0.632
Silt content	−0.446	−0.88	0.657
Clay content	−0.164	−0.735	0.451

* the correlation is significant at the 0.05 level (two-tailed).
** the correlation is significant at the 0.01 level (two-tailed).

therefore used the dye coverage of each layer as the dependent variable and the microaggregate content and plant root indexes (D > 5 mm) as the independent variable in a multiple linear regression analysis (Tables 5 and 6; Fig. 6). Before this analysis, we performed a correlation analysis on the plant root indexes (D > 5 mm) and found that the correlation between the total surface area of the plant roots (D > 5 mm) and the other indicators was very significant, and so we concluded that this index (the total surface area) could be used for the correlation analysis. From the results of the analysis, it could be seen that both the microaggregate content and total surface area of the plant roots (D > 5 mm) had an effect on the dye coverage of each layer and that the effect of the microaggregate content was the most significant. The formula used for the multiple linear regression was

$$z = -0.44158x + 0.00639y + 45.35806 \tag{3}$$

where z is the dye coverage of each layer, x is the microaggregate content, and y is the total surface area of the plant roots (D > 5 mm).

4. Discussion

4.1. Hydrological connectivity based on dye-tracing images

We used the indexes of dye-tracing images in this study to characterize the strength of the hydrological connectivity. In Fig. 5, the slope of the second straight-line segment is the steepest, the slopes of the first and third segments are smaller, and the slope of the fourth segment is the smallest. This could mean that, at the start of the experiment, before the dye began to spread, the hydrological connectivity in this area was weak. The hydrological connectivity then started to strengthen, which resulted in the dye spreading quickly through the soil. This then meant that the dye coverage was lower, and the hydrological connectivity became weaker again. The dye solution then began to diffuse into the surroundings again. Finally, there was little of the dye remaining, and the dye coverage fell to an extremely small value (Dai et al., 2020b; Zhang et al., 2017). The trends in the dye coverage values at the different plots are roughly the same. At the beginning, Pc1, had a greater coverage than the other two plots; however, it had the lowest coverage at the final stage of the experiment. This means that the upper layer of the soil at Pc1 had the weakest hydrological connectivity, whereas the middle layer (10–30 cm) here had the strongest connectivity (Zhang et al., 2018). The overall trends at the other two plots are similar. The peak value of the hydrological connectivity at Pc3 is larger than at other plots, which means that, during the dying process, there was significant fluctuation in the dye coverage at Pc3 and the heterogeneity of the hydrological connectivity of the soil was greater (Zhang et al., 2015b). For the other indexes, there is not much difference between the values at the three plots., The changes in the dyeing patterns are actually

Table 5
Analysis of the correlation between the dyeing indexes of the soil profiles and the plant root indexes.

Plant root index	Root diameter (mm)	Dye coverage of each layer	Maximum dyed depth	Length index
Total length	D < 1	0.518	−0.958	0.796
	1 < D < 3	0.429	0.838	−0.594
	3 < D < 5	0.008	0.4	−0.053
	D > 5	0.671*	−0.991	0.975
Total width	D < 1	0.218	0.863	−0.985
	1 < D < 3	0.254	−0.574	0.825
	3 < D < 5	0.202	0.003	0.348
	D > 5	0.785*	−0.593	0.838
Total surface area	D < 1	0.554	0.667	−0.886
	1 < D < 3	0.375	−0.999*	0.918
	3 < D < 5	−0.008	−0.605	0.846
	D > 5	0.831**	−0.341	0.649
Total projected area	D < 1	0.555	0.670	−0.888
	1 < D < 3	0.375	−0.999*	0.919
	3 < D < 5	−0.007	−0.610	0.849
	D > 5	0.833**	−0.337	0.646
Total volume	D < 1	0.586	0.579	−0.828
	1 < D < 3	0.357	−0.952	0.999*
	3 < D < 5	−0.079	−0.692	0.901
	D > 5	0.785*	−0.319	0.631
biomass		0.371	−0.869	0.987

* correlation is significant at the 0.05 level (two-tailed).
** correlation is significant at the 0.01 level (two-tailed).

Table 6
Results of multiple linear regression analysis.

Coefficient		Microaggregate content		Total surface area of plant roots (D > 5)		Adjusted R-squared
Value	Standard Error	Value	Standard Error	Value	Standard Error	
45.35806	32.18765	−0.44158	0.32033	0.00639	0.00198	0.68633

Table 7
Results of ANOVA analysis.

	df	SS	MS	F	Significance F
Model	2	73.74642	36.87321	9.752236	0.01302
Residual error	6	22.686	3.781001		
Total	8	96.43242			

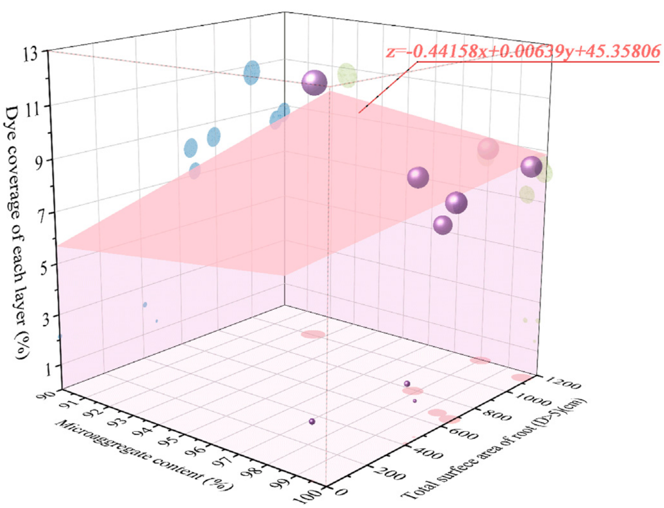

Fig. 6. Relationship among the dye coverage of each layer, the microaggregate content, and the total root surface area (D > 5 mm) (z is the dye coverage of each layer, x is the microaggregate content, and y is the total surface area of the plant roots (D > 5).).

more complicated than has been described above. However, in this study, these patterns were used only to analyze the trends in the hydrological connectivity and to explore the relationship between the changes in the hydrological connectivity and the root and soil indexes. The plots used in this study were all located near the center of a reed (*Phragmites australis*) community, and the distributions of the root system and soil within the plots were horizontally homogeneous at a small-scale. Therefore, it was reasonable to study the vertical hydrological connections under these conditions.

4.2. Hydrological connectivity and soil physical properties

Microaggregates are compound soil structures that have a diameter <0.25 mm. They are composed of organic, mineral, and other biotic materials that have combined through various physical, chemical, and biological processes (Totsche et al., 2018; Yudina et al., 2018). The stability of microaggregates allows them to remain in the soil for many years (Albalasmeh et al., 2013; Filippova et al., 2019). Their ability to repel water means that microaggregates have a great influence on hydrological process in the soil (Urbanek et al., 2007). From the results in Table 4, it can be seen the greater the microaggregate content, the smaller the dye coverage, which results in the microaggregates playing a significant role in the movement of water. It has been shown that the ability of microaggregates to repel water reduces the soil infiltration rate, making water accumulation, and surface runoff more likely. It is easy to produce a clear flow of water after infiltration into soil, which results in the moisture being unevenly distributed in the soil (Bughici and Wallach 2016; Cawson et al., 2016). Although the pore structure of microaggregates is one of main reasons for the spatial heterogeneity in the movement of water in the soil, the effects of the water repellence are clearly even more significant (Urbanek et al., 2007). Combined with the change in the dye coverage at each layer, we found that the middle layer (10–30 cm) of the soil had the largest amount of microaggregates; this resulted in the movement of water in this layer being faster. Where there were not so many microaggregates, in contrast, the water was able to diffuse through the soil. In other words, the higher the microaggregate content, the stronger the hydrological connectivity of the soil. The characteristics of microaggregates could thus be used to change the hydrological connectivity of the soil, which might then result in attempts to protect and restore wetlands being more successful. At the same time, we found that other properties of the soil (such as the porosity) had no significant correlation with the indexes of the dyeing profiles. This might be because the pores in the soil consisted of both connected and unconnected pores (Zhang et al., 2018), which meant that the probability of the water flowing to each pore was different: in general, water can flow more easily in interconnected pores.

4.3. Hydrological connectivity and plant root properties

It has been shown that there was a significant correlation between two classes of root diameter (1 mm < D < 3 mm and D > 5 mm) and the dyeing profile indexes. Other researches have shown that deeper and more complex root networks complicate the soil connection channels and that crisscrossing root networks make the soil more hydrologically connected (Scheu et al., 2003). According to the results of this study, the root system with a large diameter (D > 5 mm) had the greatest influence on the soil hydrological connectivity. This showed that the thick root squeezed the soil when growing in the soil, making the soil channel smoother. For finer roots, there was no significant relationship between soil profile indexes and the root indexes, except the negative correlation between the maximum dyed depth and the root indexes. This last result is different from those found in previous research (Zhang et al., 2015a), probably because the moisture content of the wetland soil was generally high and the capillary water in the small pores hindered the movement of the dye (Liu et al., 2020a). And the water absorption of thin roots is stronger, which would also make the dye solution difficult to flow. Another interesting phenomenon noticed in this study was that there was a significant positive correlation between the length index and the total volume of the roots (1 mm < D < 3 mm); this result was consistent with the findings of previous research. This might have been because the selection of the roots used in the experiment was not based on their vitality. However, it has been shown previously that the water flux in the rhizosphere is larger than that in other regions and that the changes in water potential in the rhizosphere are more dynamic than in other places (Johnson and Jost, 2011).

This is mainly because the secretions from the root system and the metabolites produced by the frequent renewal of the root system cause frequent significant changes in the properties of the rhizosphere (Zhang et al., 2015a). Also, root exudates can provide important energy resources. The rhizosphere contains rich microbial populations, and such large populations are rarely found in other parts of the soil (Richter et al., 2007). Thus, it can be seen that, although the root system and the soil have a certain influence on the dyeing pattern, during the actual dyeing process, the influence of the root system and soil on the hydrological processes needs to be comprehensively considered (Hinsinger et al., 2009; Jarvis et al., 2007; Johnson and Jost, 2011; Scheu et al., 2003).

4.4. Hydrological connectivity and the root–soil complex

The mechanism of the interaction between the roots and the soil is complicated. For example, because of the deep root systems and the activities of animals in soils, water is directed toward deeper soil and there is higher rate of waterflow passing through rhizosphere (Scheu et al., 2003). In addition, pores around roots are lined by hydrophobic root exudations; this can cause water to be directed to the soil zones most conducive to root development (Beven and Germann 2013; Johnson and Jost, 2011). Macropores in the soil also connect different areas within the soil and cause water flow to these areas (Mohammed et al., 2018; Watanabe and Kugisaki 2017). Root systems can only grow in pores that have diameters greater than those of the roots. Roots also produce changes in the aggregate content and porosity of the soil, and, eventually, dead root systems form new pores as they decay. (Elçi and Molz 2009; Ilstedt et al., 2007). The root system and the soil can, thus, be regarded as a whole. From the results of a multiple linear regression, we found that the soil indicators that were selected in this study play an important role in the small-scale hydrological connectivity. Therefore, it is concluded that, although the impact of plant roots may be small, it should not be ignored because plant roots have a critical effect on soil indicators such as the microaggregate content. Considering roots and soil as a whole will aid the study of hydrological processes in the soil. In this study, we neglected the influence of other soil components on the soil hydrological connectivity, which was a limitation of this study.

5. Conclusions

The new method based on the dye-tracer technology in this study could help us obtain more intuitive and accurate images of flow movement, which also helped us assess the hydrological connectivity in small scale. It was showed that the soil hydrological connectivity was significantly correlated with the microaggregate content ($r = -0.621$), total length of the roots ($D > 5$) ($r = 0.617$), total width of the roots ($D > 5$) ($r = 0.785$), total surface area of the roots ($D > 5$) (0.831), total projected area of the roots ($D > 5$) ($r = 0.833$), and total volume of the roots ($D > 5$) ($r = 0.785$). From this research, we could know that the more roots with a diameter greater than 5 cm, the stronger the hydrological connectivity, and the less the microaggregate content, the stronger the hydrological connectivity. The influence of the root–soil complex on the hydrological connections was also shown to be complex and important, with the microaggregate content being the main influencing factor. There were clearly some limitations to the experiment that was conducted: the method used was quite rough, and the indexes used cannot accurately represent the hydrological connectivity. Animals and microorganisms in the soil were also not taken into account. However, this study can be considered to have made progress in quantifying hydrological connectivity, especially small-scale hydrological connectivity.

Credit author statement

Dai Liyi: Conceptualization; Methodology; Software; Formal analysis; Investigation; Data Curation; Writing - Original Draft; Visualization. **Zhang Yinghu**: Software; Validation; Investigation; Resources; Writing - Review & Editing. **Liu Ying**: Resources; Visualization. **Xie Lumeng**: Resources; Data Curation. **Zhao Shiqiang**: Software; Formal analysis. **Zhang Zhenming**: Validation; Investigation; Writing - Review & Editing; Supervision; Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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